Bank Performance Analysis

Power BI Portfolio Project

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Project Type: Banking Performance & branch Analysis.

Tools Used: Power BI, Power Query, DAX, Excel

Year: 2025

Executive Summary

This project analyses the performance of a simulated banking dataset covering branches and monthly operations in 2025. The analysis focuses on key banking performance indicators, including deposits, withdrawals, loans issued, revenue, expenses, new accounts, and profitability.

The data was prepared and transformed using Excel and Power Query, then analysed and visualized in Power BI. An interactive dashboard was developed to identify branch-level performance, monthly trends, revenue patterns, and expense relationships.

The analysis provides a clear view of the bank's operational and financial performance and supports data-driven insights into high-performing branches, revenue generation, loan activity, and areas requiring management attention.

Project Objectives

- Evaluate **overall banking performance** using key financial and operational indicators.
- Compare **branch performance** based on revenue, expenses, loans issued, deposits, and profitability.
- Identify **monthly trends and patterns** in banking activities throughout 2025.
- Develop **an interactive Power BI dashboard** that enables users to explore performance by branch and month.

Business Questions

- Which branches generate the highest revenue and profit?
- Which branches have the highest expenses relative to their revenue?
- How do loans issue change from month to month?
- How do deposits and withdrawals vary across branches?
- Which areas of the bank performance require management attention?

Data preparation & Methodology

Data Source

The project uses a simulated banking dataset covering 2025. The dataset contains monthly records for multiple bank branches and includes key operational and financial measures such as new accounts, total deposits, total withdrawals, loans issued, revenue, and expenses.

Data Cleaning and Transformation

The dataset was initially prepared in Excel and further transformed using Power Query. The preparation process included:

- Removing duplicate records.
- Cleaning and standardising text fields.
- Trimming unnecessary spaces.
- Standardising branch names.
- Converting columns to appropriate data types.
- Ensuring the Month field was correctly formatted as a date.
- Preparing the dataset for analysis and visualisation.

Data Analysis

The cleaned dataset was loaded into Power BI for analysis. DAX measures and calculated columns were used to derive key performance indicators, including revenue, expenses, loans issued, profit, and branch-related metrics.

Dashboard Development

An interactive Power BI dashboard was developed to allow users to analyse performance by **branch and month**. Visualisations were used to highlight revenue, expenses, loans issued, profitability, and monthly trends.

Tools Used

Excel | Power Query | Power BI | DAX

Key Findings & Business Insights

Branch Performance

The analysis shows noticeable differences in performance across the bank's branches. **Aba** recorded the highest revenue among the top-performing branches, followed by **Nsukka, Gombe, Ekpoma, and Asaba**. This indicates that these branches are strong contributors to the bank's revenue generation.

The expense-ratio analysis also shows that **Owo** had the highest expense ratio among the branches displayed, followed by **Abuja, Benin City, Owerri, and Damaturu**. These branches may require closer monitoring of operating expenses to improve cost efficiency.

Revenue and Profitability

The bank generated total revenue of approximately **₦505.41 million** during the period analysed, while total expenses amounted to approximately **₦342.98 million**. This resulted in a total profit of approximately **₦162.43 million**, indicating that the bank remained profitable during the period.

The results suggest that maintaining strong revenue generation while controlling operating expenses will be important for sustaining profitability and improving overall financial performance.

Monthly Trends

The monthly analysis shows that revenue remained relatively stable from January to June, with a slight increase in February and March before declining in April and May, followed by a modest recovery in June.

Loan issuance showed greater variation during the period. Loans issued increased from January to a peak **in** February, then declined through April before recovering in May and

June. This suggests that lending activity changed considerably from month to month and should be monitored to understand the factors driving these fluctuations

Areas Requiring Management Attention

The analysis highlights several areas that may require management attention. Branches with relatively high expense ratios should be reviewed to identify opportunities for better cost control and operational efficiency.

The fluctuations in monthly loan issuance also warrant attention, particularly the decline from February to April. Management may need to investigate the factors influencing loan demand and lending activity during weaker months.

In addition, management should continue to monitor the relationship between revenue and expenses across branches to ensure that revenue growth is supported by efficient cost management and sustainable profitability.

Recommendations

• Improve cost efficiency**:** Review operating expenses in branches with high expense ratios and identify opportunities to reduce unnecessary costs.
• Strengthen high-performing branches**:** Study the practices of top revenue-generating branches and apply successful strategies where appropriate.
• Monitor lending activity**:** Investigate the decline in loan issuance after February and develop strategies to sustain lending activity during weaker months.

• Maintain profitability**:** Continue balancing revenue growth with effective expense management to protect and improve overall profitability.

Conclusion

This project demonstrates how data analysis and visualisation can be used to evaluate banking performance and support informed business decision-making. The Power BI dashboard provides an interactive view of branch and monthly performance across key financial and operational indicators.

The analysis identified differences in branch performance, revenue generation, expense efficiency, and lending activity. These findings provide management with useful information for monitoring performance, controlling costs, strengthening profitable operations, and improving lending strategies.

Overall, the project demonstrates the value of transforming raw business data into meaningful insights through Excel, Power Query, Power BI, and DAX.

Project Deliverables

- Power BI interactive dashboard
- Dashboard screenshots
- Raw dataset
- Cleaned dataset.
- Power BI `.pbix` file
- Project presentation
- Project report